

Exact Rational Stream-Dual Certificates for a Two-Bidder, Two-Item Auction

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Abstract

We study revenue maximization for two additive bidders and two heterogeneous items when all four values are independent and uniform on $[0, 1]$. The mechanism class contains every measurable, integrable, possibly randomized mechanism that is dominant-strategy incentive compatible, ex-post individually rational, and ex-post feasible. We construct an explicit rational continuous weak-duality witness in a stream-function representation. Its divergence-free, tangent-boundary correction preserves the revenue identity, while itemwise feasibility reduces the resulting objective to the integral of a pointwise maximum of polynomial virtual coefficients. A deterministic adaptive tensor-Bernstein certificate upper-bounds this continuous envelope with directed fixed-point rounding, two selective refinement levels, and exact integer accumulation. The resulting bound is

$$\text{OPT} \leq \frac{3715139591287203}{4194304000000000} = 0.8857583025186545848846435546875.$$

This is strictly below the rigorous continuous DSIC upper certificate 0.8919 reported by Jiang, Parkes, and Wang. Their paper also reports approximately 0.876 revenue for a fully strategyproof GemNet mechanism; both numbers are external benchmarks. We also give an explicit deterministic layered menu mechanism with exact revenue

$$\frac{83962078694672281756033}{9600000000000000000000} = 0.8746049864028362682920104166\dots$$

The remaining exact gap is

$$\frac{34262987107793868572569}{30720000000000000000000000} = 0.0111533161158183165926331380\dots$$

The endpoints do not match, so the unrestricted optimum remains open. The contribution is the instance-specific rational witness and explicit lower mechanism with independently reproducible exact certificates, not a new general theory of continuous auction duality.

Scope of the result. The unrestricted continuous DSIC optimum is not determined here. We prove a fully specified deterministic lower bound and an independent exact global upper bound for all randomized mechanisms. The values do not match.

1 Introduction

Optimal auction design is completely understood for a single item under the classical regularity conditions of Myerson [6]. With several items, bidder types become multidimensional and dominant-strategy incentive compatibility imposes global cross-type restrictions. Even the

canonical case of two additive bidders, two items, and independent uniform values has resisted an exact characterization [4, 8].

Computational work has produced strong feasible mechanisms and rigorous upper bounds. Affine-maximizer searches provide transparent deterministic benchmarks [8], while GemNet constructs fully strategyproof menu mechanisms after a compatibility transformation [9]. Jiang, Parkes, and Wang report approximately 0.876 revenue for that GemNet benchmark on the present instance. On the dual side, the same paper develops flow-conserving DSIC kernels and a continuous lifting theorem; its strict 50×50 uniform continuous certificate is 0.8919 [4]. These are external benchmarks. The GemNet value is a reported computational revenue, whereas 0.8919 is a rigorous upper certificate. They leave room for smaller exact endpoints whose arithmetic and continuous integration can be replayed independently.

We pursue a target-specific continuous witness rather than another grid lift. Pointwise DSIC represents each bidder’s truthful utility as a convex potential [7]. A radial integration identity produces a base vector field with the required distributional source. We then add a rational polynomial curl generated by a stream function. The correction has zero divergence and zero normal boundary component, so it changes neither the conservation law nor the revenue pairing. Ex-post feasibility bounds the paired allocation pointwise by the positive maximum of the two bidders’ field components.

The remaining problem is a four-dimensional continuous polynomial-envelope integral. We certify it using tensor Bernstein control coefficients, exact rational construction, directed fixed-point rounding, and a certificate-aware nonuniform dyadic traversal concentrated near unresolved winner-switching boxes. The certificate proves the exact upper bound $\frac{3715139591287203}{41943040000000000}$. A separate implementation reconstructs the polynomial charts and repeats the complete depth-21 base traversal plus selective depth-22 and depth-23 refinement without importing the formal verifier. As a secondary result, exact rational polytope and menu integration certify a layered lower mechanism with twenty common-fee rows, 41 bundle-pivot cells, and eight non-common item-containment cells, yielding endpoint $\frac{83962078694672281756033}{96000000000000000000000000000000}$.

Contributions. The paper makes four bounded contributions:

1. an explicit 32-parameter rational stream correction that yields a valid continuous DSIC weak-duality witness for the canonical two-bidder, two-item uniform instance;
2. an exact Bernstein-based integration certificate, including directed rounding, overflow bounds, deterministic transcripts, and an independent full-depth replay; and
3. an explicit deterministic layered taxation mechanism whose pointwise DSIC, ex-post IR, feasibility, exact revenue, and predecessor chain are checked by mutually independent rational replays; and
4. the rigorous bracket

$$\frac{83962078694672281756033}{96000000000000000000000000000000} \leq \text{OPT} \leq \frac{3715139591287203}{41943040000000000}.$$

We do not claim a general new continuous duality theorem, a one-to-one identification with every flow kernel, or global optimality of the lower mechanism. The reported Jiang–Parkes–Wang upper certificate and GemNet revenue are comparison values from the cited papers, not contributions of this work.

2 Relation to prior dual formulations

Rochet’s cyclic-monotonicity characterization turns DSIC into convexity of truthful utility and identifies allocation vectors with subgradients [7]. Continuous duality for the one-bidder multi-item problem has been developed through transport and partial-derivative formulations [2, 3].

These results motivate integration-by-parts certificates but do not directly enforce item supply across several bidders.

Multi-bidder dual formulations also predate the present work. Finite-type flow-induced virtual values arise in unified Bayesian mechanism-design duality [1]; Zuo considers both Bayesian and dominant-strategy formulations [10]; and Beckmann-style transport gives a continuous strong-duality formulation for multi-item, multi-bidder BIC auctions in interim space [5]. The latter problem uses weaker incentive and individual-rationality requirements than the pointwise DSIC and ex-post IR requirements studied here.

Most directly, Jiang, Parkes, and Wang construct nonnegative, flow-conserving DSIC deviation kernels, derive continuous weak duality, and lift coarse uniform-grid certificates to continuous type spaces [4]. Our proof shares the same high-level weak-duality architecture: conservation eliminates payments, the dual object induces virtual coefficients, and ex-post feasibility bounds virtual surplus itemwise. The representation and certification are different. We use a signed vector field with a distributional radial source plus a divergence-free tangent stream correction, and prove its validity directly against Lipschitz convex utilities. We do not assert that every such stream field is generated by a nonnegative deviation kernel, or conversely. The novelty claimed here is narrower: an explicit rational witness for one canonical DSIC instance and an exact Bernstein certificate for its continuous objective.

3 Setting and proved bounds

Write a report profile as

$$(x, y, z, w) = (v_{11}, v_{12}, v_{21}, v_{22}) \in [0, 1]^4.$$

The four coordinates are independent and uniform. A mechanism specifies jointly measurable expected allocation probabilities $x_{ij} : [0, 1]^4 \rightarrow [0, 1]$ and expected payments $p_i : [0, 1]^4 \rightarrow \mathbb{R}$, with $p_i \in L^1([0, 1]^4)$. Allocations may arise from arbitrary internal randomization; all feasibility, DSIC, and ex-post-IR inequalities concern these expectations pointwise.

Theorem 1 (Audited bracket). *For the unrestricted feasible, DSIC, ex-post-IR problem,*

$$\frac{83962078694672281756033}{9600000000000000000000} \leq \text{OPT} \leq U_{\text{stream}},$$

where

$$U_{\text{stream}} = \frac{3715139591287203}{4194304000000000} = 0.8857583025186545848846435546875.$$

For comparison, the simpler uncorrected radial field gives the weaker analytic bound

$$\begin{aligned} U_{\text{rad}} &= \frac{2}{9} + \frac{2}{3\sqrt{3}} + \frac{4}{9} \log \sqrt{3} \\ &\quad + \frac{2}{9\sqrt{3}} \left[\text{Li}_2(2 - \sqrt{3}) - \text{Li}_2(\sqrt{3} - 2) \right] = 0.920577474278749 \dots \end{aligned}$$

The lower endpoint is

$$0.8746049864028362682920104166 \dots$$

The lower bound is proved in Sections 4–6; the two upper bounds are proved in Section 7. As an external comparison, Jiang, Parkes, and Wang report a strict continuous upper certificate 0.8919 and fully strategyproof GemNet revenue approximately 0.876 for this instance [4]. Our stream bound strictly improves their printed upper decimal by the exact rational difference

$$\frac{8919}{10000} - U_{\text{stream}} = \frac{25760146312797}{4194304000000000} > 0.$$

This comparison is to their reported decimal, not a reconstruction of their underlying dual array. The 0.876 GemNet revenue is likewise a reported computational benchmark, not an exact rational certificate replayed here.

4 An exact base affine maximizer

Set

$$a = \frac{159}{250}, \quad b = \frac{91}{100}, \quad s = \frac{1137}{1000}.$$

For each feasible outcome A , let c_A and the affine score be as follows. The row number is also the tie-breaking priority.

id	allocation	score	c_A
0	seller keeps both	0	0
1	$1 \rightarrow 1$	$x - a$	a
2	$2 \rightarrow 1$	$y - a$	a
3	$\{1, 2\} \rightarrow 1$	$x + y - b$	b
4	$1 \rightarrow 2$	$z - a$	a
5	$2 \rightarrow 2$	$w - a$	a
6	$\{1, 2\} \rightarrow 2$	$z + w - b$	b
7	$1 \rightarrow 1, 2 \rightarrow 2$	$x + w - s$	s
8	$2 \rightarrow 1, 1 \rightarrow 2$	$y + z - s$	s

The base outcome $A^0(v)$ is the smallest-id maximizer of the nine scores. This defines the outcome on every equality surface.

Let

$$H_1(z, w) = \max\{0, z - a, w - a, z + w - b\}, \quad H_2(x, y) = \max\{0, x - a, y - a, x + y - b\}.$$

If $V_i(A)$ is bidder i 's reported value for the items assigned in A , the base payments are

$$p_1^0 = H_1 - V_2(A^0) + c_{A^0}, \quad p_2^0 = H_2 - V_1(A^0) + c_{A^0}.$$

Lemma 2. *The base mechanism is feasible, pointwise DSIC, and pointwise ex-post IR.*

Proof. Every listed outcome is feasible. Fix bidder 2's report. Apart from the constant H_1 , bidder 1's utility from an induced outcome A , evaluated at the true type, is

$$V_1(A) + V_2(A) - c_A - H_1.$$

Truthful reporting selects a maximizer of the expression preceding H_1 ; no report can give greater utility. The same argument applies to bidder 2. It remains valid on ties because the selected row is still a maximizer. The outcomes excluding bidder 1 attain H_1 , so the winning score is at least H_1 , and bidder 1's truthful utility is nonnegative. The symmetric argument proves bidder 2's IR. $\square$

For a bidder facing opponent report $q = (q_1, q_2)$, define

$$H(q) = \max\{0, q_1 - a, q_2 - a, q_1 + q_2 - b\}.$$

The base taxation menu has prices

$$A_0(q) = H(q) + \min\{a, s - q_2\}, \quad B_0(q) = H(q) + \min\{a, s - q_1\}, \quad C_0(q) = H(q) + b \quad (1)$$

for item 1, item 2, and the bundle, respectively; opt-out costs zero. These formulas follow directly by minimizing the outcome cost net of the opponent's value among outcomes assigning the indicated bundle.

5 The entry-surcharge mechanism

For an opponent report $q = (q_1, q_2)$, set

$$t = \max\{q_1, q_2\}, \quad \rho = \min\{q_1, q_2\}.$$

The active mechanism is the finite rational rule frozen in

`certificate/refined_item_containment_bundle_pivot_lower_bound/manifest.json`.

Its complete evaluable tables and endpoint conventions are included in that certificate directory; we summarize the three-layer construction here.

First, twenty rational Z/S half-bands add a common fee $F(q)$ to all three nonempty prices in (1). If none matches, 41 positive cells in the bundle-pivot chamber are tested using

$$u = t + \rho, \quad c = b - a = 137/500, \quad d = s - a = 501/1000, \quad r = \frac{\rho - c}{u - c - d}.$$

Those cells also add one common fee to all nonempty prices. The Z/S rows have priority on shared boundaries. Their interiors are disjoint from the bundle cells: Z support has $u \leq b$ and S support has $\rho \leq c$, whereas a bundle-cell interior has $u > b$ and $\rho > c$.

Finally, eight rows inside the two S5 half-bands add $\delta(q)$ to the singleton corresponding to the opponent's low coordinate and to the grand bundle, leaving the other singleton unchanged. Each row is left-open and right-closed in t ; at $t = 139/200$, $\delta = 0$. If maximum utility is zero, choose opt-out. At a positive tie, retain the predecessor-selected outcome whenever it remains maximizing; otherwise choose the maximizing itemwise subset with smallest id under the fixed predecessor order.

Proposition 3. *The refined surcharge mechanism is deterministic, measurable, ex-post feasible, pointwise DSIC, and pointwise ex-post IR on the entire cube, including all menu, band, cell, and item-surcharge boundaries.*

Proof. Every common fee preserves the ranking among nonempty bundles, so its selected bundle is the base bundle or opt-out. On an item row, orient the predecessor prices as

$$(A, B, C) = (d + F, t + F, t + c + F).$$

The row changes them to $(A + \delta, B, C + \delta)$, and its exact verifier checks

$$0 < \delta < A + B - C = d - c + F.$$

If the changed singleton was selected, its utility lead over the unchanged singleton is at least $A + B - C$, and hence remains positive after the change; its ranking against the bundle is unchanged. The unchanged singleton remains selected when it was selected, a bundle buyer can move only to a subset, and an opt-out cannot enter because no nonempty price falls. Thus every final bundle is an itemwise subset of its predecessor bundle, which is itself a subset of the feasible base allocation. This proves pointwise feasibility.

For each fixed opponent report the declared rule selects a utility maximizer from a taxation menu independent of the bidder's report, proving DSIC. Opt-out at price zero and the zero-utility rule prove ex-post IR. The finite rational semialgebraic tables and literal endpoint priority are Borel measurable. Numerical searches selected rational rows and fees, but no discovery file is a verifier input. $\square$

6 Exact revenue

Lemma 4 (Revenue of a four-option menu). *Suppose*

$$0 \leq A, B \leq C \leq 1, \quad C \leq A + B.$$

Let an additive type be uniform on $[0, 1]^2$. The expected payment from the four-option menu is

$$\begin{aligned} r(A, B, C) = & A(1 - A)(C - A) + B(1 - B)(C - B) \\ & + C \left[(1 - C + B)(1 - C + A) - \frac{(A + B - C)^2}{2} \right]. \end{aligned}$$

Proof. The strict item-1 region has area $(1 - A)(C - A)$, and the strict item-2 region has area $(1 - B)(C - B)$. The bundle region starts with the rectangle $[C - B, 1] \times [C - A, 1]$ and removes a right triangle of side $A + B - C$. Multiplying the three areas by their prices proves the formula. Ties have area zero for this integral; their pointwise allocation was already fixed in Proposition 3. $\square$

If the common surcharge is e , direct expansion gives

$$r(A + e, B + e, C + e) - r(A, B, C) = eG - \frac{3C}{2}e^2 - \frac{1}{2}e^3, \quad (2)$$

where

$$\begin{aligned} G = & (C - A)(1 - 2A) + (C - B)(1 - 2B) + (1 - C + A)(1 - C + B) \\ & - \frac{(A + B - C)^2}{2} - C(A + B - C). \end{aligned}$$

For a Z-band orientation the prices and cross-section length are

$$(A, B, C) = (s - t, a, b), \quad b - t,$$

and for an S-band orientation they are

$$(A, B, C) = (t, s - a, t + b - a), \quad b - a.$$

Exact integration of the twenty frozen half-bands gives

$$G_Z = \frac{113291698497241117}{10000000000000000000}, \quad G_S = \frac{2400553481931223}{3125000000000000000},$$

and therefore the common-fee gain

$$G_Z + G_S = \frac{190109409919040253}{10000000000000000000}.$$

On a bundle-pivot cell the prices are

$$(A, B, C) = (u - c, \rho + 227/1000, u).$$

The normalized map has Jacobian $u - 31/40$; including both orientations and both bidders gives symmetry factor four. Direct bivariate integration of the 41 positive cells gives

$$G_B = \frac{70946101529751}{10240000000000000000}.$$

The eight item-containment rows give

$$G_I = \frac{3963982653557301}{80000000000000000000}.$$

The base mechanism's exact expected revenue is

$$R_0 = \frac{26232089810531183}{30000000000000000}.$$

The exact verifier in `certificate/ama_lower_bound` reconstructs the $9 \cdot 4 \cdot 4 = 144$ affine/pivot cells using rational arithmetic, finds 39 full-dimensional cells, triangulates them into 16,336 rational four-simplices, checks total volume one, and integrates the affine payments. The active primary checker in `certificate/refined_item_containment_bundle_pivot_lower_bound` binds every predecessor by SHA-256, directly reconstructs the eight item-containment price regimes, checks their deletion slack and coverage, and integrates their exact symbolic polynomials. Its non-importing replay instead constructs exact rational demand polygons for those eight rows and applies exact Boole quadrature. The hash-bound sealed predecessors separately recompute the 20 common-fee rows and 41 positive bundle-pivot cells; the bundle predecessor has its own direct bivariate integration and independent scalar tensor-quadrature replay. Therefore

$$R_0 + G_Z + G_S + G_B + G_I = \frac{83962078694672281756033}{9600000000000000000000},$$

proving the lower half of Theorem 1.

The bundle-pivot, twenty-band, ten-band, and former two-rectangle certificates are retained as reproducible predecessors in their separate directories; none is the active lower endpoint.

7 Exact upper bounds for all randomized mechanisms

Fix bidder i and an opponent report on a full-measure set for which the conditional expected-payment slice is integrable. Write truthful utility as $u(v)$, expected allocation as $X(v) \in [0, 1]^2$, and expected payment as $p(v)$. Pointwise DSIC gives

$$u(v) \geq u(r) + (v - r) \cdot X(r) \quad \text{for every } v, r.$$

Thus u is convex and $X(r) \in \partial u(r)$. These inequalities and $0 \leq X_j \leq 1$ also make u Lipschitz. At every differentiability point, $X = \nabla u$.

Joint measurability and the assumed integrability of the payment, together with boundedness of $v \cdot X$, make $u = v \cdot X - p$ measurable and integrable. Thus the conditional identities below apply on this full-measure set of opponent reports and may then be integrated by Fubini.

Ex-post IR gives $u(0) \geq 0$. We do not normalize the mechanism: doing so would require an unnecessary integrability check for the opponent-dependent quantity $u(0)$.

Partition the square into the two radial charts

$$v = s(1, t) \quad \text{and} \quad v = s(t, 1), \quad 0 \leq s, t \leq 1,$$

each with Jacobian s . On a fixed ray set $q = (1, t)$ or $(t, 1)$ and $g(s) = u(sq)$. Convexity makes g absolutely continuous and $g'(s) = q \cdot X(sq)$ almost everywhere and $g(0) = u(0)$. Since $p(sq) = sg'(s) - g(s)$,

$$\int_0^1 p(sq)s \, ds = \int_0^1 \frac{3s^2 - 1}{2} g'(s) \, ds - \frac{u(0)}{2}.$$

The two charts together contribute the term $-u(0)$, which is nonpositive and may be dropped for an upper bound. Equivalently, with

$$\phi(s) = \frac{3s^2 - 1}{2s},$$

conditional expected revenue is at most the integral of $\phi(s)q \cdot X(sq)$ against the chart measure $s ds dt$.

Apply this identity to both bidders and then integrate over opponent reports. For every profile and item j , feasibility gives $X_{1j} + X_{2j} \leq 1$, with both allocations nonnegative. Hence

$$\sum_i \phi(s_i) q_{ij} X_{ij} \leq \max\{0, \phi(s_1) q_{1j}, \phi(s_2) q_{2j}\}.$$

This pointwise relaxation is valid for arbitrary expected allocations and therefore for arbitrary internal randomization.

For completeness, the remaining scalar expectation can be evaluated exactly. Under the two radial charts, a uniform type can be generated by taking S with density $2s$, an independent fair choice of which coordinate is maximal, and an independent $T \sim U[0, 1]$. For one fixed item its radial coordinate therefore has distribution

$$Q \sim \frac{1}{2} \delta_1 + \frac{1}{2} U[0, 1].$$

Set $Y = \max\{0, \phi(S)\}$, independently of Q . For two bidders let $a = \max(Y_1, Y_2)$ and $b = \min(Y_1, Y_2)$. Conditioning on the four atom/uniform cases for Q_1, Q_2 gives

$$\mathbb{E}_{Q_1, Q_2} \max\{Y_1 Q_1, Y_2 Q_2\} = \frac{3}{4} a + \frac{b^2}{6a},$$

where b^2/a is defined as zero when $a = 0$. Since the two items are symmetric,

$$U_{\text{rad}} = \frac{3}{2} \mathbb{E}[a] + \frac{1}{3} \mathbb{E}\left[\frac{b^2}{a}\right]. \quad (3)$$

Put $r = 1/\sqrt{3}$. The function ϕ is increasing from zero to one on $[r, 1]$, while $Y = 0$ below r . The larger of two independent S 's has density $4s^3$, hence

$$\mathbb{E}[a] = \int_r^1 4s^3 \phi(s) ds = \frac{8}{15} + \frac{4}{45\sqrt{3}}.$$

Ordering the two S 's gives the second moment as the explicit integral

$$\mathbb{E}\left[\frac{b^2}{a}\right] = 8 \int_r^1 \frac{s}{\phi(s)} \left(\int_r^s t \phi(t)^2 dt \right) ds.$$

The inner integral is

$$J(s) = \frac{9}{16} s^4 - \frac{3}{4} s^2 + \frac{1}{4} \log s + \frac{3}{16} + \frac{1}{8} \log 3,$$

so the last display equals

$$16 \int_r^1 \frac{s^2 J(s)}{3s^2 - 1} ds.$$

Set $x = \sqrt{3}s$. The cancellation of the $\log 3$ terms in J reduces this integral to

$$\frac{1}{3\sqrt{3}} \int_1^{\sqrt{3}} x^2 (x^2 - 3) dx + \frac{4}{3\sqrt{3}} \int_1^{\sqrt{3}} \frac{x^2 \log x}{x^2 - 1} dx.$$

Use $x^2/(x^2 - 1) = 1 + 1/(x^2 - 1)$. For the only non-elementary term, the substitution $y = (x - 1)/(x + 1)$ has upper endpoint $d = 2 - \sqrt{3}$ and gives

$$\int_1^{\sqrt{3}} \frac{\log x}{x^2 - 1} dx = \frac{1}{2} \int_0^d \frac{\log(1 + y) - \log(1 - y)}{y} dy = \frac{1}{2} [\text{Li}_2(d) - \text{Li}_2(-d)],$$

where

$$\frac{d}{dz} \text{Li}_2(z) = -\frac{\log(1 - z)}{z}$$

was used in the last equality. The remaining polynomial and $\int \log x \, dx$ terms then give

$$-\frac{26}{15} + \frac{8}{5\sqrt{3}} + \frac{4}{3} \log \sqrt{3} + \frac{2}{3\sqrt{3}} [\text{Li}_2(d) - \text{Li}_2(-d)].$$

Substitution in (3) gives

$$U_{\text{rad}} = \frac{2}{9} + \frac{2}{3\sqrt{3}} + \frac{4}{9} \log \sqrt{3} + \frac{2}{9\sqrt{3}} [\text{Li}_2(2 - \sqrt{3}) - \text{Li}_2(\sqrt{3} - 2)].$$

This proves the analytic benchmark $\text{OPT} \leq U_{\text{rad}}$. We next improve it by an exact divergence-free correction.

7.1 A rational continuous stream certificate

For one bidder with own type (x, y) , opponent type (z, w) , and $r = \max(x, y)$, write the radial vector field as

$$F^0(x, y) = \frac{3r^2 - 1}{2r^2}(x, y).$$

Define $F^0(0, 0) = 0$; this value is immaterial. Although the field is singular at the origin, $|F^0| = O(1/r)$, so it is locally integrable in two dimensions. Its classical divergence away from the origin is 3, its outer boundary flux is 2, and its inward flux around a puncture at the boundary corner $(0, 0)$ tends to 1. Because that corner is not an interior point of the open square, we use the following punctured-domain identity rather than a bare distributional-divergence shorthand. Applying Green's formula outside the square $[0, \varepsilon]^2$ and then letting $\varepsilon \downarrow 0$ gives, for every Lipschitz truthful utility,

$$\int F^0 \cdot \nabla u = \int_{x=1} u + \int_{y=1} u - 3 \int_{[0,1]^2} u + u(0).$$

Indeed, the puncture flux tends to one and Lipschitz continuity replaces the trace of u there by $u(0)$ with vanishing error. This is equivalently the integration-by-parts identity

$$\int_{[0,1]^2} p(v) \, dv = \int_{[0,1]^2} F^0(v) \cdot X(v) \, dv - u(0).$$

We now add an opponent-dependent curl. For an exponent $e = (e_0, e_1, e_2, e_3)$, put

$$m_e = x^{e_0} y^{e_1} z^{e_2} w^{e_3}, \quad \sigma(e) = (e_1, e_0, e_3, e_2).$$

Let E be the lexicographically smaller member of every nontrivial σ -pair of total degree at most four. Define

$$h(x, y, z, w) = \sum_{e \in E} \theta_e (m_e - m_{\sigma(e)}), \quad \psi = x(1-x)y(1-y)h, \quad G = (\partial_y \psi, -\partial_x \psi).$$

The thirty-two rational coefficients are listed below in the exact basis order used by the verifier. A convex quasi-Monte Carlo search was used only to discover them; the proof defines the field

by the following rational numbers and does not trust the numerical optimization.

e	θ_e	e	θ_e
(0, 0, 0, 1)	-2995385/993184	(0, 0, 0, 2)	2337131/676164
(0, 0, 0, 3)	5724621/964727	(0, 0, 0, 4)	-4591777/714341
(0, 0, 1, 2)	-834567/846478	(0, 0, 1, 3)	6291553/970203
(0, 1, 0, 0)	2762675/446201	(0, 1, 0, 1)	206964/961681
(0, 1, 0, 2)	-6090207/979156	(0, 1, 0, 3)	6862090/913753
(0, 1, 1, 0)	-5206994/854575	(0, 1, 1, 1)	515889/117356
(0, 1, 1, 2)	-3346363/843816	(0, 1, 2, 0)	-125263/465506
(0, 1, 2, 1)	3005651/895053	(0, 1, 3, 0)	-1043238/468269
(0, 2, 0, 0)	1660919/994155	(0, 2, 0, 1)	1276309/943222
(0, 2, 0, 2)	-2075136/255347	(0, 2, 1, 0)	404816/59269
(0, 2, 1, 1)	250988/352269	(0, 2, 2, 0)	3825425/994154
(0, 3, 0, 0)	-5383501/354845	(0, 3, 0, 1)	3312081/940771
(0, 3, 1, 0)	-4885471/759831	(0, 4, 0, 0)	7648954/974695
(1, 1, 0, 1)	-603823/508896	(1, 1, 0, 2)	417332/188831
(1, 2, 0, 0)	17884893/955196	(1, 2, 0, 1)	992961/785407
(1, 2, 1, 0)	2761171/712822	(1, 3, 0, 0)	-3841999/611370

For every fixed opponent report, $\text{div}_{x,y} G = 0$. Moreover $G \cdot n = 0$ on each side of the own-type square: on a vertical side $\partial_y \psi = 0$, and on a horizontal side $-\partial_x \psi = 0$. Therefore adding G changes neither the conservation law nor the boundary term. Indeed, the polynomial field is bounded and the Lipschitz utility belongs to $W^{1,\infty}$, so weak Green integration gives $\int G \cdot \nabla u = 0$ on almost every opponent slice; its joint polynomial dependence then permits Fubini. With bidder 2 receiving the same field after swapping (x, y) with (z, w) , set $F_i = F_i^0 + G_i$. For every randomized feasible allocation,

$$\sum_i F_{ij} X_{ij} \leq \max\{0, F_{1j}, F_{2j}\},$$

and consequently

$$\text{OPT} \leq \int_{[0,1]^4} \sum_{j=1}^2 \max\{0, F_{1j}(v), F_{2j}(v)\} dv. \quad (4)$$

No implementability or weak-monotonicity property of the pointwise maximizing allocation is used here.

It remains to bound the integral in (4) exactly. Split each bidder square into (s, st) and (st, s) . In each of the four chart pairs, multiplication by the Jacobian $s_1 s_2$ turns both item-1 competitors into exact rational polynomials in (s_1, t_1, s_2, t_2) . The antisymmetry of h under $(x, y, z, w) \mapsto (y, x, w, z)$ exchanges the two field components; invariance of the uniform distribution therefore makes the item-2 integral identical. The two charts overlap only on the diagonal and at the origin, which are null sets.

The certificate converts each polynomial and their exact difference to tensor Bernstein form. A polynomial lies in the convex hull of its Bernstein control coefficients, and its box integral is the box volume times their mean. Every exact initial control c is stored as an integer $m = \lfloor 10^9 c \rfloor$ with error radius one. A degree- n de Casteljau half-box split makes n floor averages, so the rigorous radius update is $E \leftarrow E + n$. Lower bounds for a competitor and for the exact competitor difference certify a fixed winner; otherwise the maximum upper control gives a valid supremum charge.

The tree uses the maximum-variation axis above the final four levels of a depth-21 base partition and exact one-step child-bound lookahead in those final four levels. Each unresolved

depth-21 leaf is split once along the best exact child-bound axis if and only if that split saves at least one integer accumulator unit; otherwise its depth-21 charge is retained. Each unresolved child of an accepted depth-22 split is then tested independently: its best depth-23 split is accepted if and only if it saves at least one further unit, and otherwise its depth-22 charge is retained. Taking the minimum of several valid upper bounds is valid. All contributions are accumulated over the common depth-23 dyadic denominator.

The manifest-driven verifier reconstructs the polynomials and reports

boxes fixed before unresolved depth-21 leaves :	404804,
unresolved depth-21 leaves :	462796,
leaves refined to depth 22 :	391618,
depth-21 leaves retained :	71178,
fixed depth-22 children :	173198,
unresolved depth-22 children :	610038,
depth-22 children refined :	609951,
depth-22 children retained :	87,
fixed depth-23 children :	249438,
unresolved depth-23 children :	970464,
integer accumulator :	3715139591287203.

After the four chart pairs and the item-symmetry factor two, this gives

$$\int \sum_j \max\{0, F_{1j}, F_{2j}\} \leq \frac{2 \cdot 3715139591287203}{10^9 2^{23}} = \boxed{\frac{3715139591287203}{4194304000000000}}.$$

This proves the upper half of Theorem 1. The verifier and its clean transcript are in the following certificate directory:

`certificate/continuous_stream_degree4_two_level_nonuniform_upper_bound/`

7.2 Reproducibility and independent replay

The publication archive separates the proof inputs from the numerical search that found them. The trusted upper-bound directory contains the rational coefficient manifest, polynomial constructor, exact verifier, non-importing replay, and captured transcripts; the root SHA-256 manifest covers these files. The verifier checks all manifest expectations and fails closed on any mismatch. Before fixed-point conversion, every polynomial and Bernstein coefficient is constructed with `fractions.Fraction`. The largest axis degree is eight, the maximum initial absolute control is 1475346733, and the maximum observed directed-rounding radius is 181 fixed-point units. Signed array operations are checked against $2^{63} - 1$; child charges and the global accumulator use arbitrary-precision Python integers.

The non-importing iterative replay independently reconstructs the basis, chart maps, polynomial arithmetic, Bernstein conversion, subdivision, winner tests, selective-refinement rule, and coverage accounting. Both implementations visit 3,738,334 retained-tree nodes and reproduce all refinement counts, with

$$\text{accumulator} = 3715139591287203, \quad \text{coverage} = 33554432/33554432.$$

These replays audit the finite arithmetic; the surrounding continuous weak-duality argument is the analytic proof above.

The lower-bound chain reconstructs the 144 affine/pivot cells, twenty common-fee rows, 41 positive bundle-pivot cells, and eight item-containment rows from rational manifests. Its independent implementations use scalar rational arithmetic, exact forward-difference degree checks,

exact Boole quadrature, and, for the final non-common perturbation, exact demand-polygon clipping. The root reproduction command reruns all publication-facing checks, compares their structured outputs with the theorem endpoints, and verifies the archive hash manifest. No optimizer output is read anywhere on the trusted proof path.

8 Discussion and open problem

The certified lower and upper values are unequal. The corrected stream bound still relaxes cross-type convex-gradient compatibility after using DSIC, so equality would require a pointwise field maximizer that need not itself be cyclically monotone. Conversely, the lower mechanism is one explicit deterministic menu construction; no valid theorem reduces the allowed class of randomized mechanisms to this family. Consequently neither the mechanism nor its revenue is asserted to be globally optimal.

The stream certificate is an exact continuous weak-duality witness and narrows the previously reported rigorous bracket, but it is not complementary to the present primal mechanism. Its improvement came from combining the same frozen degree-four rational witness with deeper, two-level nonuniform, certificate-aware local Bernstein refinement. The lower improvement came from adding deletion-compatible common-fee, bundle-pivot, and item-containment layers to the same exactly verified affine base. Sampled objectives were discovery evidence only. The exact residual width is

$$U_{\text{stream}} - \frac{83962078694672281756033}{9600000000000000000000} = \frac{34262987107793868572569}{3072000000000000000000} \\ = 0.0111533161158183165926331380 \dots$$

A useful next step is to improve the witness jointly with its local certificate geometry, or to retain more of the convex-gradient restrictions discarded by the itemwise pointwise maximum. Closing the exact primal–dual equality remains open.

Data, code, and materials availability

The accompanying archive contains all exact certificate data, deterministic verification code, transcripts, environment information, and SHA-256 manifests needed to reproduce the stated bounds. Manuscript, certificate data, and other textual materials are released under CC BY 4.0. Verification software is separately released under the MIT License. Discovery-only experiments are segregated from the trusted proof artifacts.

AI-assisted tools declaration

This manuscript was completed with the assistance of OpenAI GPT-5.6 Sol.

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